Displacement and velocity

1

a $v\left(t\right)=12t$

b i Yes ii No iii No iv Yes

c $48\ \mathrm{m/s}$

2

a $v\left(t\right)=9t^{2}-8t$

b $20\ \mathrm{m/s}$

3

a $v\left(t\right)=6t+5$

b $29\ \mathrm{m/s}$

4

a $v\left(t\right)=\dfrac{9}{\sqrt{t}}$

b $3\ \mathrm{m/s}$

5

a $67.5\ \mathrm{m/s}$

b $v\left(t\right)=\dfrac{3t^{2}}{2}+6t$

c $67.5\ \mathrm{m/s}$

d $t=2$ s

6

a $0\ \mathrm{m/s}$

b $(-9.8x+11.27)\ \mathrm{m/s}$

c $11.27\ \mathrm{m/s}$

d $5.635\ \mathrm{m/s}$

e $0\ \mathrm{m/s}$

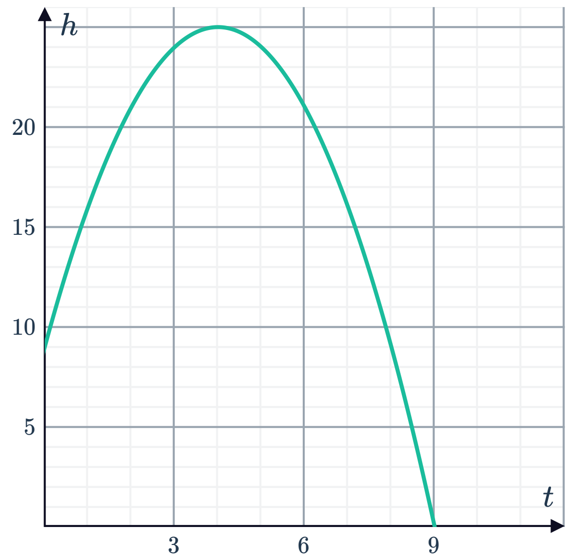
7

a $9\ \mathrm{m}$

b $h=25\ \mathrm{m}$

c $t=9$ s

d



8

a $h = 16\text{ m}$

b $t = 6\text{ s}$

c $\frac{dh}{dt} = -32\text{ m/s}$

9

a -4 m

b $v(t) = 6t^2 - 14t + 3$

c 3 m/s

d 1 m/s

10

a 9 m

b $v(t) = 32t + 24$

c 24 m/s

d 24 m/s

11

a 0 m

b $t = 4\text{ s}$

c 40 m

12

a 0 m

b $v(t) = -4t - 20$

c -20 m/s

d 20 m/s

13

a $\frac{ds}{dt} = 12 - 9.4t$

b $\frac{d^2s}{dt^2} = -9.4\text{ m/s}^2$

14

a $\frac{dh}{dt} = -13\text{ m/s}$

b $\frac{d^2h}{dt^2} = -10\text{ m/s}^2$

15 $a = 1, b = 8$

Acceleration

16

a $v(t) = 4t - 4$

b $t = 1\text{ s}$

c $a(t) = 4$

17

a $v(t) = 2t - 7$

b -7 m/s

c $a(t) = 2$

d 2 m/s^2

e $t = 5\text{ s}$

f $t = 5, 2\text{ s}$

18

a $\frac{d^2s}{dt^2} = t + \frac{1}{2}$

b $\frac{1}{2}\text{ m/s}^2$

c $\frac{33}{2}\text{ m/s}^2$

19

a $\dfrac{d^2s}{dt^2} = 12t^2 + 2$ **b** $2\ \mathrm{m/s^2}$ **c** $110\ \mathrm{m/s^2}$

20

a $a = 38 - 12t$ **b** $-10\ \mathrm{m/s^2}$

c The particle is slowing down.

21

a $t = 4\ \mathrm{s}$

b $0\ \mathrm{m/s}$

c $t = 2\ \mathrm{s}$

d $12\ \mathrm{m/s}$

e The velocity is constant. The particle is neither speeding up or slowing down.

22

a $t = \dfrac{3}{2}\ \mathrm{m/s}$

b $a(2) = 20\ \mathrm{m/s^2}$

c The particle starts moving to the right, but eventually comes to rest. After instantaneously coming to rest, it then continues to move to the right.

23 $v = -\dfrac{25}{2}\ \mathrm{m/s}$

24 $x = -12\ \mathrm{m}$

25

a $t = 8\ \mathrm{s}, 3\ \mathrm{s}$ **b** $a(t) = 2t - 11$ **c** $-7\ \mathrm{m/s^2}$

26

a $t = \dfrac{8}{5}\ \mathrm{s}, 3\ \mathrm{s}$ **b** $a(t) = 10t - 23$ **c** $7\ \mathrm{m/s^2}$

27

a $t = 2$ **b** $x''(t) = -6t - 2$

c $x''(2) = -14\ \mathrm{m/s^2}$ **d** $4\ \mathrm{m/s}$

28

a $d' = 1.72t - 0.0252t^2$ **b** $t = 68\ \mathrm{s}$

c $1335\ \mathrm{m}$ **d** $d' = 29\ \mathrm{m/s}$

29

a $\frac{9}{10}$ s

b $s'(t) = u + at$

c $t = -\frac{u}{a}$

d $a = -\frac{50}{9}$ m/s^2

Graphs of displacement, velocity and acceleration

30

a 0 m

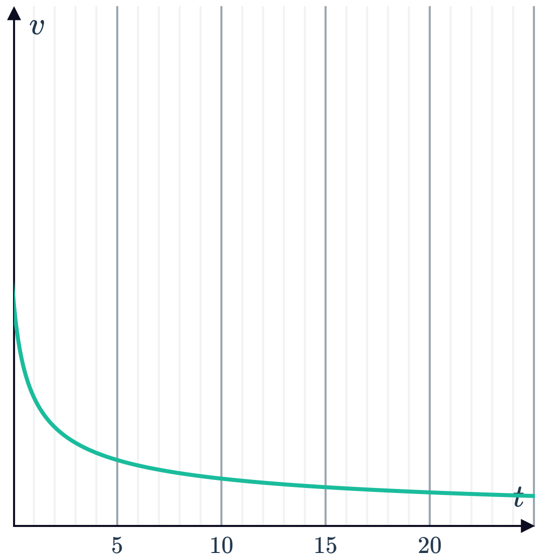
b 30 m

31

a -6 m

b 6 m

32



33

a $3, 5, 10$

b $0 < t < 4, t > 8$

c $4, 8$

34

a $6, 11$

b $6 < t < 11$

35

a
i -2 m ii 8 m

b -3 m/s

c 2 m/s

36

a 2s

b $0 < t < 2$

c 4 m

d $x = -t(t-4)$

e 4 s

f $\frac{dx}{dt} = -2t + 4$

g -2 m/s

37

a 9 m **b** 6 s, 15 s **c** To the left **d** 11 s

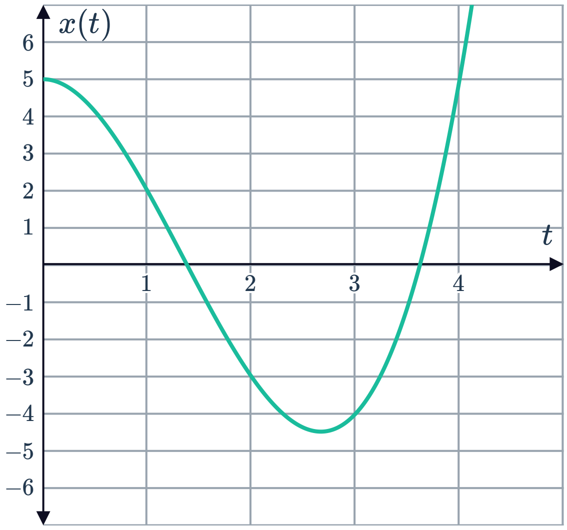
e $-\frac{8}{5}$ m/s

38

a $v(t) = 3t^2 - 8t$

b $t = 0$ s, $\frac{8}{3}$ s

c



d $\frac{256}{27}$ units

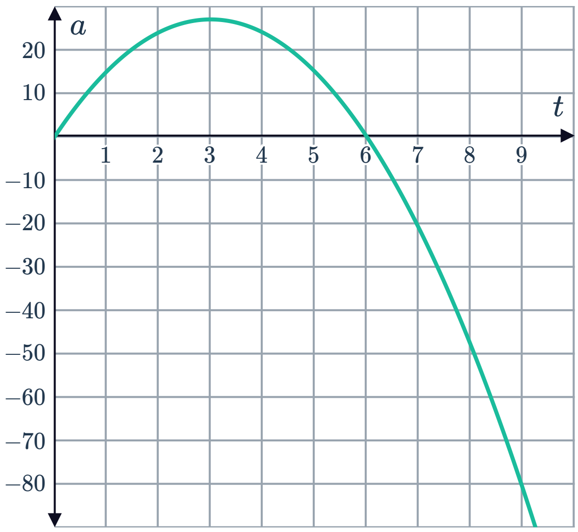
39

a $a = 18t - 3t^2$

b $t = 6$ s

c 108

d



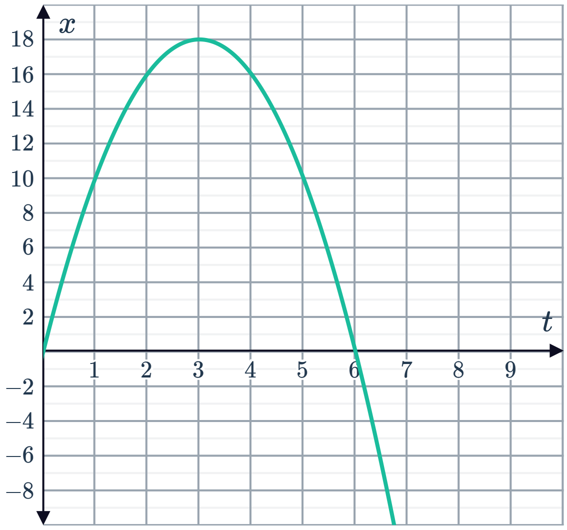
e 81

40

a $v = 12 - 4t$

b $t = 3\text{ s}$

c



d -54 m

e 90 m

41

a $x'(t) = 3t^2 - 18t + 24$

b $x''(t) = 6t - 18$

c

i

$+$	$t < 2$
0	$t = 2$
$-$	$2 < t < 4$
0	$t = 4$
$+$	$t > 4$

ii

$-$	$t < 3$
0	$t = 3$
$+$	$t > 3$

d $2\text{ s}, 4\text{ s}$

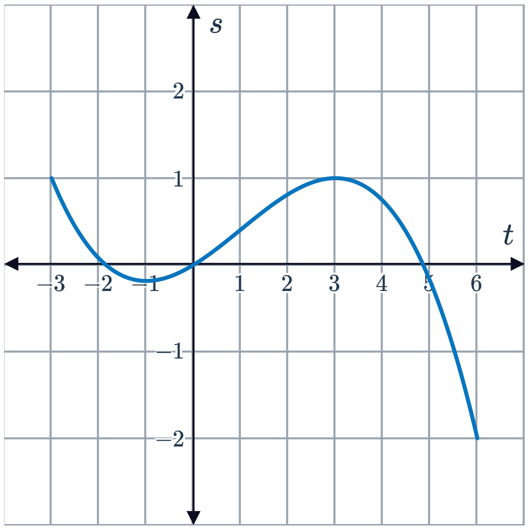
e $t < 2, 3 < t < 4$

f $t < 3$

g 28 m

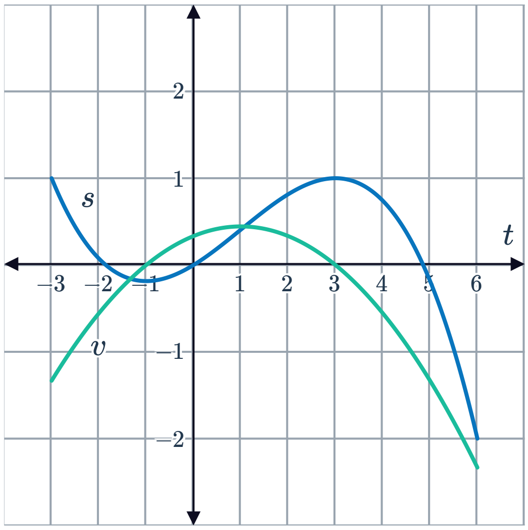
42

a



b $\frac{ds}{dt} = \frac{1}{3} + \frac{2}{9}t - \frac{1}{9}t^2$

c



d 4 s

e $-\frac{5}{27} \text{ m}$

f 1 m

g $\frac{8}{27} \text{ m/s}$

Displacement and acceleration from velocity

43

a

i $0 \leq t \leq 10$

ii No values of t.

b

i No values of t.

ii $0 \leq t \leq 10$

c

i $0 \leq t \leq 4, 8 \leq t \leq 10$

ii $4 \leq t \leq 8$

d

i $7 \leq t \leq 10$

ii $0 \leq t \leq 5$

44

a

i 11 units

ii 19 units

b

i 6 units

ii 18 units

c

i -3 units (or 3 units to the left)

ii 15 units

d

i 1.75 units

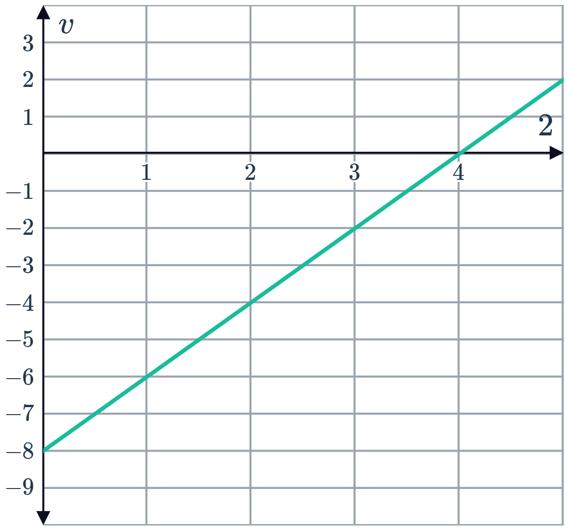
ii 19.75 units

45 Example answers:

- Because both are equal to the area of the triangle above the horizontal axis from $t = 0$ to $t = 6$.
- Because the particle only moved in one direction during the first six seconds.

46

a

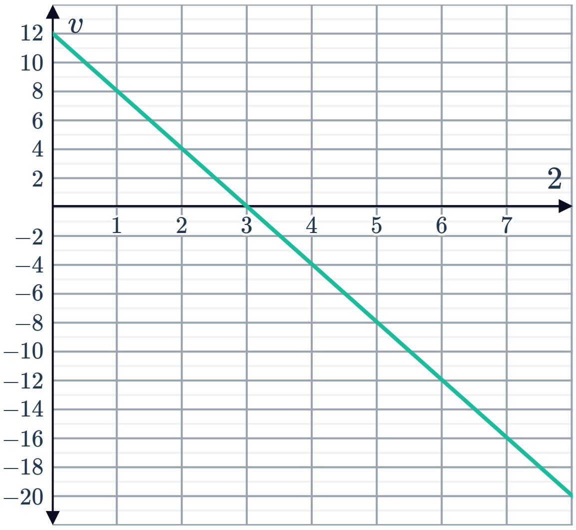


b -15 units or 15 units to the left.

c 17 units.

47

a

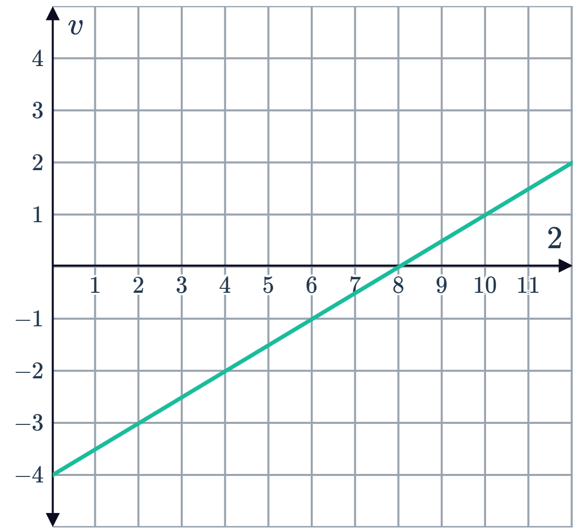


b -10 units or 10 units to the left.

c 26 units.

48

a



b -3 units or 3 units to the left.

c 5 units.